

Spatiotemporal variation of power law exponent on the use of wind energy

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HIGHLIGHTS

- Assess the spatiotemporal variation of the power law exponent (α) in global wind energy assessments.
- The power law exponent α values vary with geography, seasons, diurnal cycles, latitudes, land and ocean.
- The power law is generally applicable and has important implications for wind energy resource assessments and turbine design.

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ABSTRACT

The power law, a common method in wind energy assessments, often assumes a global power law exponent (α) of 1/7, potentially leading to inaccuracies in estimating wind conditions and energy potential. This study uses high-frequency measurements from eight wind towers to assess the power law's reliability and demonstrates its general applicability, except in complex topography. The α value is found to differ significantly from 1/7 in validation sites, yet it can be reliably used in ERA5 data based on regional verification. To evaluate the global spatiotemporal variation of α , we analyze the hourly α from 1980 to 2022 using ERA5 wind speed data at 10 m and 100 m. Our findings reveal varying α value with time and geography, with particular disparities between land and ocean. Additionally, we observe changes related to diurnal cycles, seasons, and latitudes. This analysis of α value carries significant implications for a global-scale wind energy resource assessment, site selection, and turbine design.

1. Introduction

The Paris Agreement calls for an accelerated reduction in global greenhouse gas emissions in the background of global warming. Conversion from traditional energy production into renewable energy production is an essential improvement towards the zero-carbon emission goal [1]. Wind energy currently shares around 7% of nowadays global total electricity needs and is projected to generate one-third of total electricity needs by 2050 [2]. It is imperative to assess global wind energy generation. The generation of wind energy through wind turbines generally follows that the generated wind energy is proportional to the cube (third power) of the wind speed [3]. Accordingly, the wind energy generation can benefit from proper wind turbine site selections and

design considering average wind speed at turbine heights. Turbine heights usually range from 50 to 150 m above ground. Yet available wind speed data are mostly at the height of 10 m. Problems emerge when extrapolating the observed near-surface wind speeds to the turbine height wind speeds used for wind energy assessment.

Several methods can be used to obtain wind speeds at turbine heights. For example, the Global Wind Atlases provide high spatial resolution wind resource assessments by integrating meteorological models; however, they are only able to provide wind speeds at fixed heights [4]. As wind turbine technology has advanced, turbines with hub heights of 200 m or higher have emerged, and these fixed-height global wind datasets are often insufficient to meet the demands of wind energy project development. On the other hand, we can

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extrapolate the upper-level wind speeds based on logarithm, power law formulas or others. The logarithm law is a calculation method to take into account the influence of atmospheric stability on the surface wind profile [5,6]. However, the parameters that this model needs to consider, such as the boundary layer height cannot be easily obtained, so the wind power law shown in Eq. (1) as a quick and efficient approach has been widely used for wind speeds extrapolation and wind energy generation assessment [7]. Moreover, the wind power law commonly describes the wind speed patterns in the boundary layer and only needs the near surface wind speed as input [8].

$$\frac{u_1}{u_2} = \left(\frac{h_1}{h_2} \right)^\alpha \quad (1)$$

The wind power law is widely utilized to convert near-surface wind speed at h_1 (u_1) to wind speed at hub-height h_2 (u_2), where α is the empirically derived wind power law exponent that varies with the atmospheric stability, surface roughness, wind speed and height interval [7]. For this reason, an accurate α value determines whether the wind speed at the wind turbines' height can be accurately calculated, which is critical for developing wind energy.

However, wind power law is oversimplified in previous studies by using a constant 1/7 value worldwide as the wind power law exponent [9–13]. The α value of 1/7 and its characteristics are debated especially in recent studies [14–17]. In realistic circumstances, the observed power law exponent α exhibits spatiotemporal variation regarding the conditions including atmospheric stability and surface roughness [18–20]. Besides, the 1/7 α value has poor representativeness in many regions such as Southeast Asia [21,22], North America [23], Europe [24] and offshore areas [25] as well as temporal variation. There are various studies using in-situ measurements [19], lidars and sodars [26,27] data to analyze the characteristics of the power law exponent α of the wind profile, which claim that α value with spatial and temporal variation can be more accurate in estimating wind energy than using fixed value. However, these early studies using measurement data are plagued by spatial and temporal data limitations, making global analysis difficult. There are also remote sensing data [28,29] and reanalysis data [30,31] that are used for wind speed extrapolation on a large regional scale, though the data from satellites are normally limited over the ocean. With the ascendance of offshore wind turbines, assessments of offshore wind energy potential are becoming increasingly critical as well as the onshore wind energy. While Jung's study has analyzed the properties of α using global land data, it ignored the broader ocean and compute the global grids only on the seasonal temporal scale [32]. Therefore, it is urgent and necessary for a global assessment of spatiotemporal variation of α on covering both land and ocean for more accurate and efficient global wind energy estimation, given the development of climatology and the availability of big data till the most current, which has not been done globally before.

In this study, we use wind tower data as a regional validation to test whether the wind profile follows the power law and calculate the exponent α for each tower using a statistical model. The measured α values are then compared to the commonly used 1/7 value and reanalysis values to determine the errors as a previous study shows that reanalysis may underestimate α value [33], but rarely other comparisons. In our study, the wind tower data are accessible only in China, they can be used as a regional stand-in for the real data to assess whether α values follow the power law and how close they are to 1/7. Meanwhile, we use global reanalysis data to investigate the spatiotemporal features, diurnal, seasonal and interannual variation, and trends of α at a global scale. We will better understand the global spatial and temporal variation of α value in order to make up for many previous studies limited to the regional scale and have identified issues that will merit further investigation. In general, shorter time-span data allow for more precise analysis of the instantaneous wind shear characteristics, just as many studies mentioned above use short-term high-frequency observation

data to analyze. However, it is also appropriate to utilize long-term data analysis because such data has a considerable degree of accuracy and stability [34]. We recommend incorporating spatiotemporally differentiated α value in wind energy estimation research at any location on earth, particularly when hub-height wind speed data is unavailable on a large time and spatial scale. This approach can effectively replace the simplistic use of 1/7. Finally, we publish our research results as open data. All the findings will help to improve wind energy assessments.

In the subsequent analysis, the data sources and computational methodologies employed are first elaborated upon in Section 2. Section 3 provides verification and comparison of the power law relationship through the use of measured data from eight towers, reanalysis data, the conventional 1/7 exponent value. Then examines the global spatiotemporal variation and trends in the power law exponent α , including diurnal, seasonal, and interannual variability. Section 4 compares the results in this study with other published findings and discusses the limitations of the ERA5 results. Finally, Section 5 offers concluding remarks and summary comments.

2. Method

2.1. Data source

This study utilized two data sources. One is the ERA5 global reanalysis hourly averaged data (<https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=form>) [35]. ERA5 data is the fifth-generation data provided by the European Centre for Medium-Range Weather Forecasts (ECMWF) including the global climate and weather data for the past several decades. ECMWF owns one of the largest supercomputers facilitating and archiving meteorological data in the world, assuring the reliability of the data used in this study. Some studies have shown that the near-surface wind speeds from the ERA5 reanalysis data are closer to the measured data than several other reanalysis data products [36]. The ERA5 reanalysis data are widely used as the data source for many studies on wind speeds and the wind power law [32,37,38]. Here we used ERA5 global hourly averaged reanalysis data from 1980 to 2022 to investigate the spatial and temporal variation of the power law exponent on a global scale. This $0.25^\circ \times 0.25^\circ$ resolution gridded dataset provides the u -component and v -component of wind speeds at 10 m and 100 m.

Another data source is the anemometer wind tower data provided by Mingyang Smart Energy Company (<http://www.myse.com.cn/>). There are a total of eight towers positioned across China, each of which is capable of measuring wind speed at multiple heights. Table 1 depicts the information for each tower. The data range from over a year to more than four years, and it is continuously updated every ten minutes

Table 1
The information of the 8-wind tower measured data.

Wind tower	Location	Terrain	Time of data	Height of data
Hunan	25°N 111°E	Mountain	2013.1–2013.12	10/30/50/60/ 70 m
Xinjiang	43°N 88°E	Dessert	2018.1–2018.12	10/30/50/70 m
Jiangxi	26°N 114°E	Mountain	2016.1–2016.12	10/30/50/70/ 80/90 m
Shanxi	37°N 113°E	Mountain	2019.5–2020.4	10/30/50/70/ 80/90 m
Anhui	33°N 116°E	Plain	2018.11–2019.10	10/30/50/70/ 120/140 m
Jiangsu_Land	32°N 119°E	Plain	2019.6–2020.5	10/60/80/ 100/120/140 m
Qinghai	36°N 96°E	Dessert	2013.1–2013.12	10/30/50/70 m
Jiangsu_Seaside	32°N 120°E	Plain	2014.1–2014.12	10/30/50/70/ 80/90/100 m

without any interruptions. The wind speeds in one year which the periods range from 2011 to 2020 are chosen for each tower to fit the α value. It is important to note that the measured wind speeds of eight towers are at several different heights range from 10 m to 140 m above ground level. Each tower contains at least four different heights, because of data resource constraints but also provides as much as possible wind speed data calculations at higher heights. The sites' location, heights, and topography vary greatly, which are depicted in Fig. 1. These wind towers range in altitudes from below 100 m to over 3000 m. The satellite images surrounding the towers are obtained from Google Earth, allowing the topography around the towers to be approximated. These towers can be found in various terrains, including plains, deserts and mountains, including inland and coastal areas. It is worth noting that the data used in previous studies which we have mentioned before are rarely as high in the specification as the anemometer wind towers data used in this study, whether in terms of the diverse topography, high temporal resolution, or spatial span.

2.2. Power law

The wind power law relationship is described as Eq. (1). We have made a simple transformation of the way it is written to calculate the power law exponent as follows:

$$\alpha = \frac{\ln\left(\frac{u_1}{u_2}\right)}{\ln\left(\frac{h_1}{h_2}\right)} \quad (2)$$

and the equation is related to the wind speed at two heights. As a result, the α value for a given wind turbine can be computed by either solving regressions or directly using data from two heights. It should be noted that we used the square root of u -component squared plus v -component squared of the 10 m and 100 m wind speeds respectively to directly calculate the α value for the ERA5 reanalysis wind speed data:

$$u_{10} = \sqrt{(uwind_{10})^2 + (vwind_{10})^2} \quad (3)$$

$$u_{100} = \sqrt{(uwind_{100})^2 + (vwind_{100})^2} \quad (4)$$

To increase the reliability of the α value for the anemometer wind tower data, we solved the nonlinear regression model using wind speed data collected at as much as possible distinct heights in each of the measured data to calculate the tower α value. The regression model is based on the least-squared approach and the transformation of Eq. (1):

$$u = u_{10} \times \left(\frac{h}{10}\right)^{\alpha} + b \quad (5)$$

where h is the height correspond to Table 1 except 10 m, u is the corresponding wind speed and u_{10} is the wind speed at 10 m, b is the error

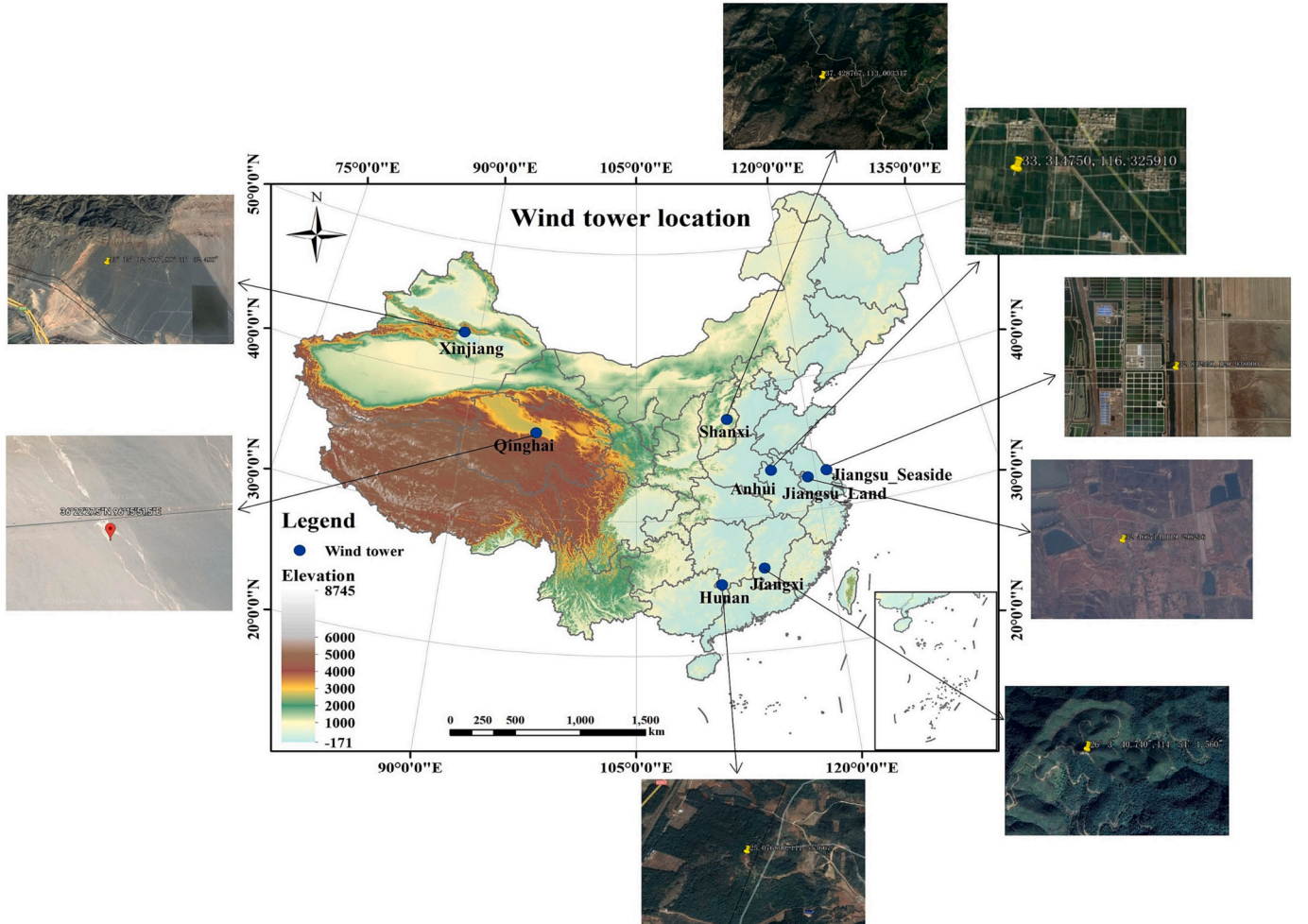


Fig. 1. The location of eight wind towers. The towers are distributed in China, and their images are obtained from the screenshots on Google Earth.

term according to statistical principle. This equation designates u as the dependent variable because the source of errors lies in the deviations of wind speed, making our regression procedure aim to minimize the deviation of u , which is also consistent with the physical mechanism of power law. We use the coefficient of determination (denoted R^2) as the proportion of the variance of dependent variable that can be well explained by the regressions and p -value from t -test to see the significance of relationship (Hypothesis: there is no such nonlinear relationship). R^2 closer to 1 and p -value smaller than 0.05 stands for a good regression result. When calculating the α for the towers, we also utilize eq. (6) based on least squares method. The distinction lies in transforming Eq. (1) into a linear form. We consider that linear regression offers greater robustness, and this formulation comprehensively captures the relationship among all height pairs. It's important to note that when introducing an error term, it alters the physical mechanism of the power law (it becomes an adjustment for α , not u). Consequently, we assume the intercept of the fitted straight line is 0. As Eq. (6) lacks a statistical error term, we use it solely as a reference in this case.

$$\alpha = \frac{\sum_{i=2,j=1,h1>h2}^n \left(\ln \left(\frac{h_{1,i}}{h_{2,j}} \right) \right) \left(\ln \left(\frac{u_{1,i}}{u_{2,j}} \right) \right)}{\sum_{i=2,j=1,h1>h2}^n \left(\ln \left(\frac{h_{1,i}}{h_{2,j}} \right) \right)^2} \quad (6)$$

In Fig. S1, the hourly α value results obtained by the two fitting equations are slightly different. Alternatively, it is reasonable that the transformation of the regression equation calculates very different results, the research can also be conducted by swapping the roles of the independent and dependent variables, and the outcome is determined based on the equation with the smaller deviation [39].

2.3. Validation of measured data

For the comparison of reanalysis data and wind tower data, we compare the errors among hourly reanalysis α value, measured α value and 1/7. The comparison of the errors of α value is defined as the following equation:

$$ratio = \frac{|\alpha_t - \alpha_e|}{|\alpha_t - 1/7|} \quad (7)$$

where α_t and α_e are the α values from wind tower data and ERA5 reanalysis data respectively.

3. Result

3.1. Validation of power law

We assume the wind speeds at various heights conform to power law. However, the wind speed in this study may not be consistent with that assumption [22,40,41]. As a test of whether the wind speeds at different heights conform to the power law distribution, we used the anemometer wind tower data to determine the α value. If the validation results are not aligned with the power law, we will need to consider whether alternative equations, such as the log law, should be used to calculate wind speeds at various heights.

The validation of the power law exponent is used for the measured

data from the eight wind towers (Table 1). First, Gap test, Syntax test and Gross range test are used for the quality control of the tower data in order to make sure the data ensure that the data are time-continuous, within the normal range of wind tower monitoring and can be used for calculation through coding [42]. Then derive hourly α values from regression analysis of wind speed data at several distinct heights. Meanwhile, calculate the R^2 and p -value of each fitted α value to estimate whether each tower is suitable to get α value using the power law. Table 2 shows the percentage of hourly R^2 larger than 0.5 and p -value less than 0.05 among all data in one year of each tower (Table S1 based on Eq. (6)). The results demonstrate that most of the hourly values display ideal fitting results and significant relationship. While tower Xinjiang, Jiangxi, and Qinghai are not fitted significantly using the power law for some time according to the p -value. It might be connected to the terrain where the tower is situated. Studies have shown that surface roughness and mountain topography significantly impact wind speed characteristics at various heights as the atmospheric stability is weaker, which result in wind speeds at different heights and directions to fluctuate to varying degrees at different times [43,44]. As a result, based on the collected data, the wind profile most comply with the power law. It indicates that the further analysis will have some degree of reliability.

Fig. 2 depicts the one-year monthly and diurnal mean α results of eight towers. The α value between each tower varies greatly, the range is about 0.4. Towers Xinjiang, Jiangxi, and Qinghai have lower α values than other five towers, which are less than 1/7 in most monthly and yearly values. The fluctuations of monthly mean α value in each tower are less than the fluctuation range among different towers. There is relatively significant diurnal variation in the 8 towers. Except for tower Xinjiang, Jiangxi and Anhui, the diurnal variation follows the characteristic that α values are lower in the daytime and higher at night. Fig. S2 based on linear regression depicts similar characteristics but generally larger values. Generally, the result shows that the α value has spatio-temporal variation and significantly different from 1/7, which requires more accurate estimation.

3.2. Compare ERA5 α values and 1/7 against in-situ measurements

We extracted the gridded data with corresponding space and time from the ERA5 reanalysis global gridded data for the eight anemometer wind tower locations (Fig. 1) and calculate the hourly α value. It is very important that the ERA5 data results can show the fluctuation of α value. Fig. 3 exhibits the mean value of diurnal and monthly fluctuations in tower Jiangsu Seaside. Comparing the ERA5 data with the measured data, the variation of the α value between day and night and between months can be well displayed (i.e., α value is lower in the day and higher at night), and the reflection of such fluctuations probably cause the 1/7 value a large error at some time.

There is still some discrepancy between the calculated results of the measured data and the reanalyzed data. To understand whether such a discrepancy is more favorable or not than uniformly using 1/7, Eq. (7) from Section 2.3 is used to verify the measured data (Fig. 4 & S3). Assuming that the fitted α from the measured data to be the accurate value, as long as the ratio is less than 1, we believe that the result derived from ERA5 is closer to the measured value compared to the 1/7. From the original hourly data, only three towers have a ratio less than 1 for

Table 2

Validation of power law in 8 towers. The first row refers to the percentage of determination coefficient R^2 which calculated from the fitted α value that larger than 0.5 among all the hourly data in one year. The second row is the percentage of p -value of the nonlinear fitting that smaller than 0.05 among all the hourly data in one year. The third row is the averaged value of the error term.

Tower	Hunan	Xinjiang	Jiangxi	Shanxi	Anhui	Jiangsu_Land	Qinghai	Jiangsu_Seaside
($R^2 > 0.5$) %	89.02	91.52	81.83	86.91	84.08	92.25	85.74	89.82
($p < 0.05$) %	71.76	29.06	64.03	74.62	77.06	87.05	33.65	85.28
Averaged b	0.92	0.15	-0.27	1.72	0.08	0.38	0.13	0.48

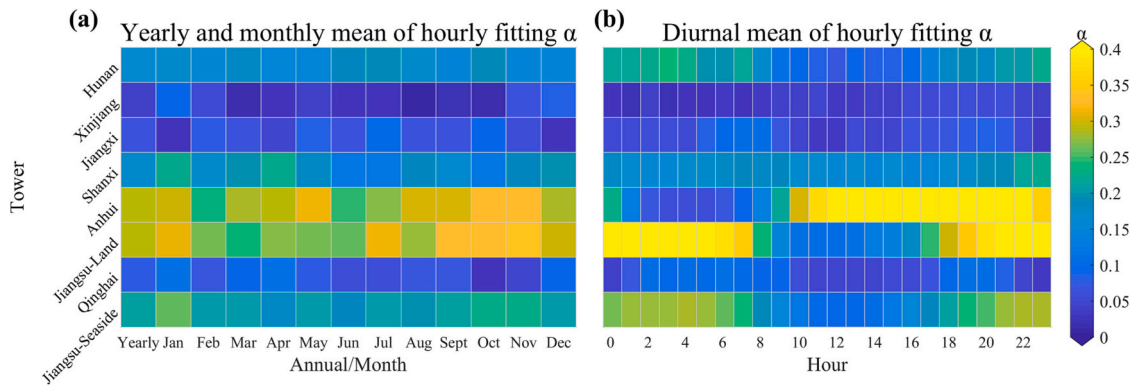


Fig. 2. Mean α value of eight towers. (a) is annual and monthly mean of hourly α value. (b) is the diurnal mean of hourly α value.

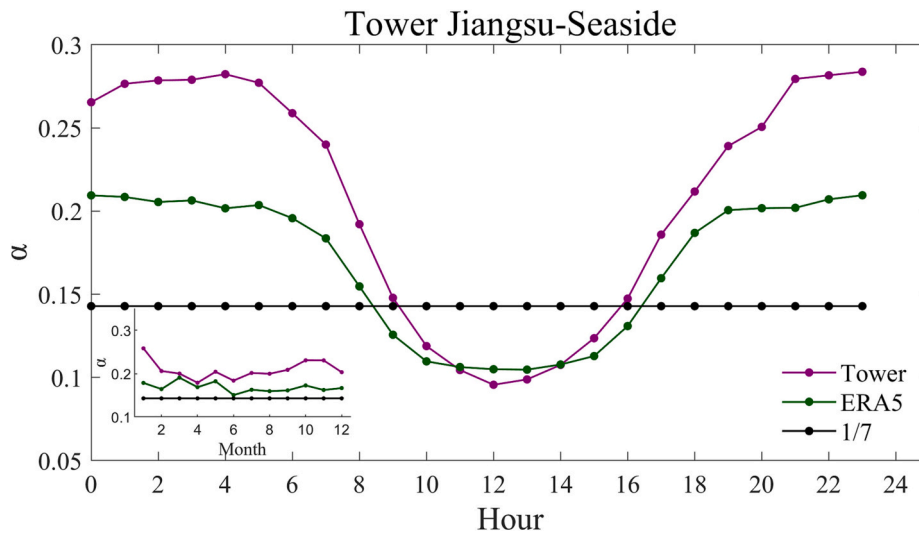


Fig. 3. An example of analysis in tower Jiangsu-Seaside. The major figure shows the diurnal variation of α value calculated from measured and reanalysis data. The small figure shows the monthly variation. Two data are compared with $1/7$.

more than half of the time, which is not an ideal result. When the hourly data is averaged into monthly data and then calculate the ratio, there are four towers where all the data are within 1, and over 50% of the ratios less than 1 in two towers. Exceptions are the tower Hunan and Jiangxi, possibly due to the wind speed data do not comply with the power law that we have proved or the nonlinear model is overparameterized (refer to the results of linear model). Overall, the calculation error of using the reanalysis data of ERA5 in wind energy production as a calculation basis is smaller than that of uniformly using $1/7$, illustrating that using ERA5 data are valuable for reference.

3.3. Spatial pattern of mean annual α value

To make the spatiotemporal variation of α value analysis be examined on larger spatial and temporal scales, the global gridded wind speed reanalysis data are used for the subsequent analyses.

Fig. 5 illustrates the multi-year mean α value from 1980 to 2022, derived from the 10 m and 100 m wind speeds. The annual values vary from the winter and summer values, represented by January and July, respectively, and the spatial variation are also shown as well. Most land areas have α values higher than $1/7$, while most ocean areas have α values lower than $1/7$, meaning there is a big difference between land and ocean, and only a few regions have values very close to $1/7$. There are a few regions in the low latitude ocean area where the α value is negative, indicating that the 10 m wind speed is greater than the 100 m wind speed due to the smooth surface of the ocean. A region with a

greater α value indicates that the upper-level wind speed is greater than the lower-level wind speeds. These are regions where we must concentrate our efforts, as higher wind speeds allow wind turbines to produce more wind energy. Additionally, there are some regions with abnormal values. For instance, in northwestern China, the α values of a few long-strip regions are negative, which are similar to Jung's result [32]. We find that these abnormal regions are mainly caused by the relatively small wind speed at 100 m and large near-surface wind, which has similar patterns in other studies [45]. This could be related to the desert climate or the sudden increase in elevation that have influence on surface roughness [46]. The locations also make sense, as the abnormal regions are located in a basin surrounded by mountains, frequently exist large near-surface wind speed at the direction of the mountain gap [47]. For example, when the prevailing wind direction is consistent with the direction of a canyon, the narrow space can accelerate the 10 m wind speed, while the wind speed at upper-level has little effect, which has a time pattern of fluctuation. The relevant study also considers smaller α value in mountainous regions with stronger mixing of atmospheric turbulence [41]. However, some studies also find that there may be biases when applying ERA5 data to mountainous and coastal regions [48]. The average α value in the ocean increases as it moves from the equator to the poles, and the average α value in the land area increases in both the Northern and Southern Hemispheres during their respective winters. The difference in α values on land is mainly caused by surface roughness, which leads to obvious regional variation. In the ocean, however, surface roughness is very low, so the difference in α is mainly

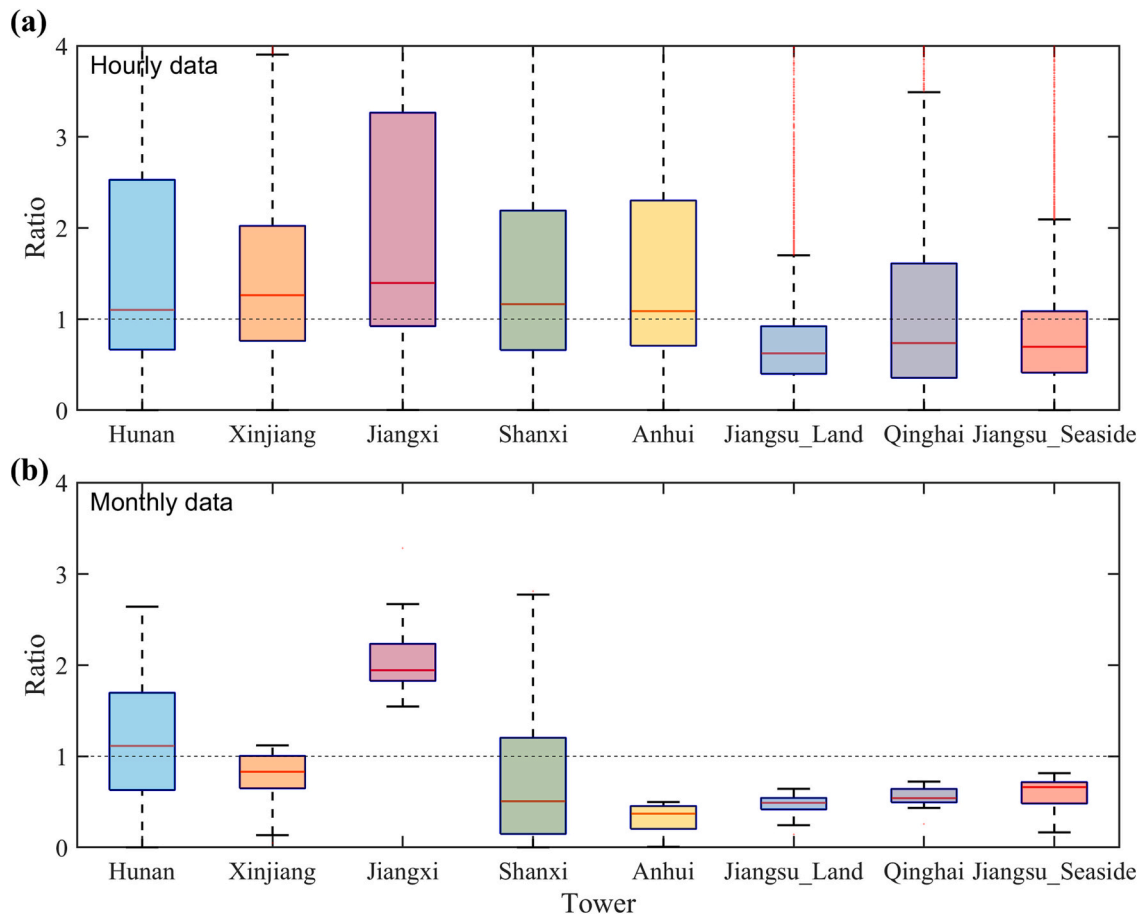


Fig. 4. Comparison of reanalysis data and 1/7 with measured data. (a). The ratio is calculated from the original hourly value in one year. (b). The ratio is calculated from the monthly mean value in one year.

affected by atmospheric stability. At the same location, low temperature enhances atmospheric stability and reduces vertical convection. This results in a more pronounced wind profile and a larger α value. Therefore, in the ocean, α values are larger at higher latitudes where the temperature is lower [26].

To correlate with the installation of wind turbines, we also calculated the multi-year average α value in different countries, which facilitates more accurate and straightforward calculations for assessing the wind power generated by individual wind turbines especially in various small-area and flat countries. Table 3 displays the α value for some major countries and regions, which is more convenient to use than gridded data. From the standpoint of wind turbines, we placed greater emphasis on countries with larger markets, as variation in α value can result in greater numerical differences in the calculation of wind power in these nations. According to WWEA's country ranking for the global wind turbine market in 2021 (<https://wwindea.org/world-market-for-wind-power-saw-another-record-year-in-2021-973-gigawatt-of-new-capacity-added/>), China is ranked in the first place with an average α value of 0.189, and the second-ranked United States has an average value of 0.208. They are both greater than 1/7, which means that when 1/7 is used to calculate upper-level wind speeds and wind power, the capacity of the two countries with the most wind turbines are underestimated. Average α values for the first ten countries range between 0.18 and 0.25, indicating that the 1/7 parameter calculation method would probably result in a significant underestimation when calculating wind power for countries with a large installed wind turbine capacity. These errors may be minimized by using the data in Table 3 rather than 1/7 for each nation.

We can also use the multi-year average α value which distinguish

various climate types when determining the upper-level wind speeds in a region (Table 4). Most of these α values are larger than 1/7. The results indicate that climate types with warmer temperatures approximately have higher values. When people obtain a certain α value, they can refer to the data in the two tables to judge the relative level of the used value and the average value in a region.

3.4. Diurnal variability

As there exists difference in the pattern of daytime and nighttime wind speeds, the diurnal variation of the multi-year mean α value should be analyzed. This will also provide a reference for the variation in power generation of wind turbines during the day and night. Due to the large difference in the surface roughness, the characteristics of the ocean and land need to be distinguished. Fig. 6 shows the hourly α values for global ocean and land averages. The global characteristic for diurnal variation is that α value is lower during the day than at night. This is mainly due to the large fluctuation of day and night on land, with a variation of about 0.1, greater than 0.2 at night and reduced to about 0.1 during the day. While the variation in the ocean is relatively small and stable at about 0.07 as the ocean's near-surface boundary environment is more stable. This is because the surface is warmed by solar radiation during the day, creating unstable conditions. At night, a stable boundary layer forms as vertical mixing is suppressed by a radiative inversion at the surface. The large heat capacity of the ocean causes the sea surface temperature to change little due to solar radiation, resulting in a strong atmospheric stability and small diurnal variation in α value [26]. Past studies have also found that the diurnal heating and cooling cycle of air above the ground affects the diurnal variation of the α value on land, as well as

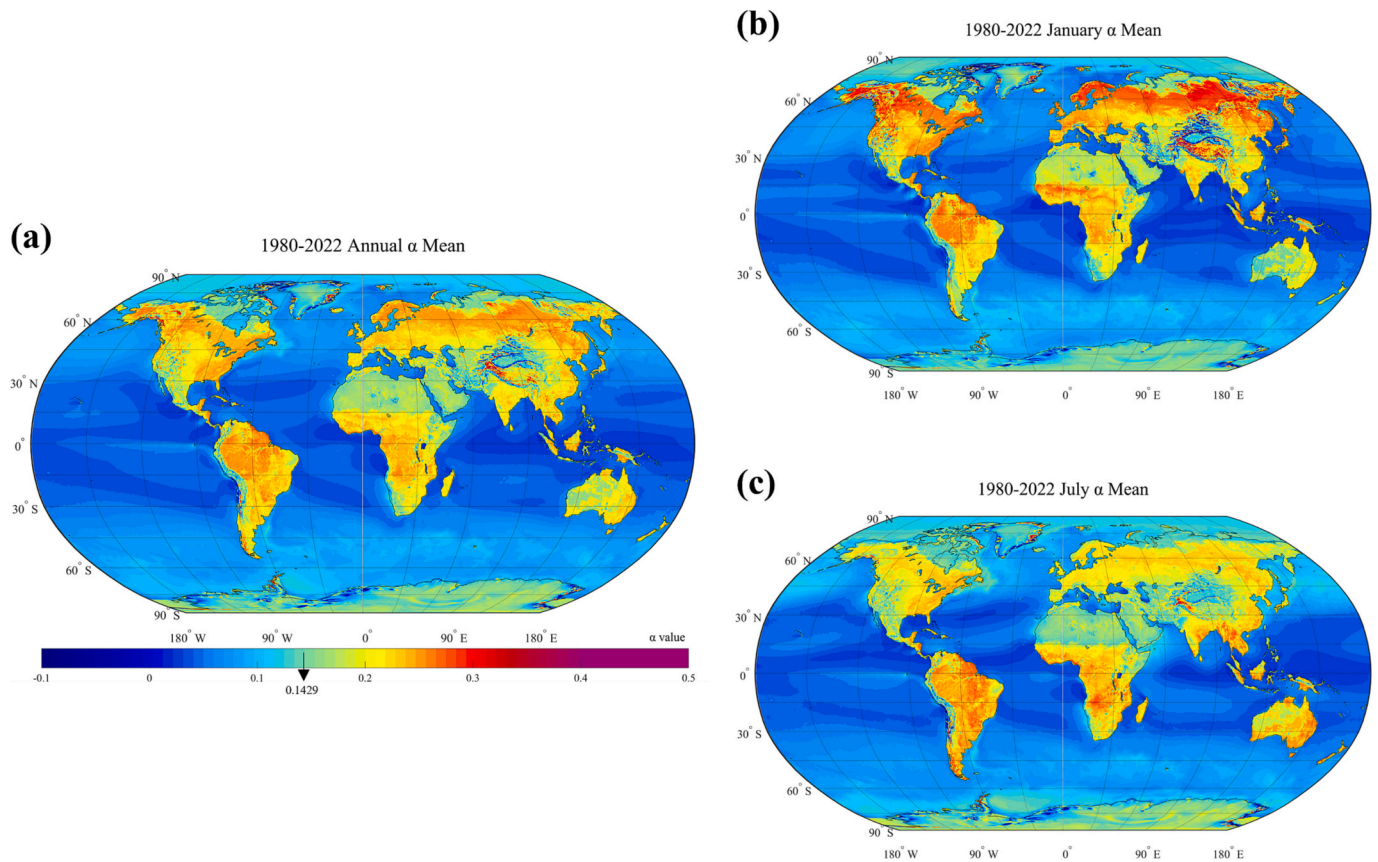


Fig. 5. Spatial pattern of multi-year average α value calculated from annual, January, and July data on a global scale. (a) The multi-year average α value of the 43 years. (b) The multi-year average α value in January of the 43 years. (c) The multi-year average α value in July of the 43 years.

similar fluctuation characteristics [19]. Result means that using averages over land without taking into account diurnal variation will overestimate daytime upper-level wind speeds and underestimate nighttime wind speeds.

3.5. Seasonal variability

In addition to spatial variation, temporal variation are important for upper-level wind speed estimation at different periods. For example, Fig. 7 shows small fluctuations in monthly average values when separating land and ocean areas in the Northern and Southern Hemispheres. The trough and peak monthly average α values in the land indicate that α values have opposite fluctuations between the Northern and Southern Hemispheres as their seasons are opposite, but the correspondence between the opposite peak and trough values in the Northern and Southern Hemispheres in the ocean is slightly misaligned. The fluctuation range is approximately 0.01 in each figure except Northern Hemisphere for larger fluctuation, from which we can see opposite seasonal variation between the land and ocean in each hemisphere that α values are higher in winter on land while higher in summer in the ocean. These characteristics are similar to previous studies, except that studies at regional sites exhibit larger fluctuations [26]. The patterns are because of changes in the atmospheric stability due to temperature changes which similar to Fig. 6, as well as the monsoon effects. This means that when judging the land as a whole, using winter's α value to calculate upper-level wind speeds will get larger values than using summer's α value, while opposite for the calculation in the ocean.

By distinguishing the spatial area, we can observe both seasonal variation and spatial variation. Fig. 8 demonstrates that α value varies with latitudes but with opposite seasonal variation characteristics over land and ocean. In the land area, the α value decreases from the equator

to the poles, particularly small above 80°N , while it increases in the ocean, particularly large between 80°S and 90°S . The α values from 60°N to 60°S on land are approximately 0.2 (Fig. 8a), whereas the values from 30°N to 30°S in the ocean are approximately 0.05 (Fig. 8b). In addition, land and ocean regions close to the poles have values closer to $1/7$. In the two figures, the seasonal variability reflected in each latitude region is not as significant as spatial variation across regions distributed by latitude.

3.6. Interannual variability

If the α value is highly unstable in the utilization of wind turbines, we cannot be sure whether the α value we calculated is suitable for use because of the great variability. This indicates that the α value at a given time is frequently quite different from the average calculated from 1980 to 2022 data. Therefore, we must examine the extent to which the α value varies over time. Here, we normalized the interannual variability of the α value using standard deviation (STD) derived from the hourly value over 43 years. Fig. 9 depicts the annual mean, winter, and summer STD characteristics using the annual average, January, and July data, respectively. Generally, higher STD values on land are typically associated with complex topography. For instance, the areas surrounding the Cordillera Mountains in America and the Qinghai-Xizang Plateau in Asia. Larger variation in the α value between years may result from wind speeds varying with average elevation and terrains in mountainous regions. Because the wind speeds in different directions are affected by the complex topography in different degrees, such as hills and valleys can channel winds or block airflow, altering seasonal variation of local wind conditions. Therefore, the impacts may differ on an annual basis due to changes in wind flow, resulting in variability in wind shear and profiles represented by α [49]. Besides, the wind speed profile may not conform

Table 3National multi-year average α value from maximum to minimum for 172 major countries and regions.

Countries	α value	Countries	α value	Countries	α value	Countries	α value
Afghanistan	0.159	Egypt	0.170	Liberia	0.206	Saudi Arabia	0.160
Albania	0.146	El Salvador	0.158	Libya	0.165	Senegal	0.205
Algeria	0.166	Equatorial Guinea	0.199	Lithuania	0.230	Sierra Leone	0.201
Angola	0.220	Eritrea	0.167	Luxembourg	0.227	Slovakia	0.225
Antarctica	0.143	Estonia	0.217	Macedonia	0.185	Slovenia	0.221
Argentina	0.206	Ethiopia	0.177	Madagascar	0.205	Solomon Islands	0.099
Armenia	0.158	Fiji	0.158	Malawi	0.206	Somalia	0.158
Australia	0.196	Finland	0.242	Malaysia	0.192	Somaliland	0.156
Austria	0.215	France	0.221	Mali	0.188	South Africa	0.188
Azerbaijan	0.162	Gabon	0.225	Mauritania	0.168	South Korea	0.209
Bangladesh	0.208	Gambia	0.209	Mexico	0.172	South Sudan	0.219
Belarus	0.222	Georgia	0.165	Moldova	0.235	Spain	0.199
Belgium	0.224	Germany	0.233	Mongolia	0.173	Sri Lanka	0.186
Belize	0.236	Ghana	0.214	Montenegro	0.183	Sudan	0.187
Benin	0.229	Greece	0.155	Morocco	0.159	Suriname	0.251
Bhutan	0.132	Guatemala	0.187	Mozambique	0.218	Swaziland	0.208
Bolivia	0.211	Guinea	0.217	Myanmar	0.199	Sweden	0.244
Bosnia and Herzegovina	0.201	Guinea Bissau	0.174	Namibia	0.168	Switzerland	0.193
Botswana	0.207	Guyana	0.253	Nepal	0.136	Syria	0.167
Brazil	0.234	Haiti	0.103	Netherlands	0.204	Taiwan	0.163
Brunei	0.170	Honduras	0.213	New Zealand	0.194	Tajikistan	0.219
Bulgaria	0.207	Hungary	0.232	Nicaragua	0.215	Thailand	0.217
Burkina Faso	0.227	Iceland	0.190	Niger	0.179	The Bahamas	0.068
Burundi	0.167	India	0.201	Nigeria	0.222	Togo	0.227
Cambodia	0.217	Indonesia	0.186	North Korea	0.202	Trinidad and Tobago	0.106
Cameroon	0.219	Iran	0.145	Northern Cyprus	0.189	Tunisia	0.178
Canada	0.190	Iraq	0.169	Norway	0.206	Turkey	0.174
Central African Republic	0.229	Ireland	0.227	Oman	0.144	Turkmenistan	0.163
Chad	0.186	Israel	0.223	Pakistan	0.162	Uganda	0.174
Chile	0.152	Italy	0.166	Palestine	0.171	Ukraine	0.221
China	0.189	Ivory Coast	0.228	Panama	0.182	United Arab Emirates	0.151
Colombia	0.214	Jamaica	0.146	Papua New Guinea	0.185	United Kingdom	0.205
Costa Rica	0.156	Japan	0.197	Paraguay	0.234	United Republic of Tanzania	0.203
Croatia	0.203	Jordan	0.159	Peru	0.171	United States of America	0.208
Cuba	0.177	Kazakhstan	0.193	Philippines	0.157	Uruguay	0.213
Cyprus	0.137	Kenya	0.183	Poland	0.228	Uzbekistan	0.166
Czech Republic	0.236	Kosovo	0.208	Portugal	0.189	Vanuatu	0.130
Democratic Republic of the Congo	0.231	Kuwait	0.159	Qatar	0.145	Venezuela	0.217
Denmark	0.183	Kyrgyzstan	0.154	Republic of Congo	0.238	Vietnam	0.214
Djibouti	0.161	Laos	0.231	Republic of Serbia	0.221	Western Sahara	0.152
Dominican Republic	0.184	Latvia	0.229	Romania	0.218	Yemen	0.135
East Timor	0.155	Lebanon	0.144	Russia	0.215	Zambia	0.230
Ecuador	0.178	Lesotho	0.176	Rwanda	0.167	Zimbabwe	0.215

to the power law [50]. Similar STD patterns also appear in Jung's study as the STD are generally small [32]. A few higher STD values exist in the polar regions, but these areas are rarely suitable for the construction of wind turbines. Other regions with smaller STD also exhibit smaller interannual variation in α value, indicating that these regions have a higher fundamental implication. Further research can concentrate on most of the global regions with α values close to zero, where the analysis results will be more stable and useful.

3.7. Trend of α value

The multi-year global average α value is lower than 0.1, with the average α value of roughly 0.168 for global land and 0.064 for ocean in Fig. 10. The interannual fluctuation of α value is not obvious compared with diurnal and monthly temporal variation and spatial variation, so here we focused more on the trend. Fig. 10 displays only weak trends in the global, land and ocean mean values, and the trends are not statistically significant refer to p -value larger than 0.05, which stands for there is no obvious trend when the global mean values are considered. Although there is no significant trend when calculating the results of the entire land or ocean, we need to know if there is a clear trend in some parts of the world. Fig. 11 shows the trends by distinguishing the annual averages, January values and July values. The trend values are indicated by the slopes derived from linear fitting over the 43 years. Trend values in most parts of the globe are distributed in $\pm 5 \times 10^{-4}$, especially on

land where areas with increasing trends alternate with areas with decreasing trends. The black dot in the figure shows that the fitted trend's p -value is less than 0.05, defined as significant (based on t -test). Generally, the global gridded trends are not obvious in most regions, which means that for wind energy assessment, it is reliable to use the α value in a certain past year for future estimations as it does not change significantly.

4. Discussion

4.1. ERA5 α value and other studies' results

Previous research used the global reanalysis wind speed dataset to analyze the power law exponent α value and the influence of wind speed on wind energy [32]. Still, their analyses are restrained only to the land area. In this study, we use the global data and compare the land α values to be larger than the ocean α values, while the values over the ocean are smaller than 1/7. The obvious difference from our research is that we first combined the measured data with the ERA5 data for analysis, which is verified by the real value. This makes it more reliable when using ERA5 data for global analysis. Similar analyzes in Jung's study to ours are presented in Figs. 6 and 10 of this study. From the view of the distribution characteristics of α values on land, Jung's results are very similar to ours. Judging from the standard deviation of the α values, the larger standard deviations of the two studies both appear in places with

Table 4
Multi-year average α value in different climate types according to the Koppen climate classification.

Climate type	α value
ET Polar-Tundra	0.141
EF Polar-Frost	0.141
Aw Tropical-Savanna	0.219
BSh Arid-Steppe-Hot	0.199
BWh Arid-Desert-Hot	0.166
Af Tropical-Rainforest	0.213
Dsa Cold-Dry_Summer-Hot_Summer	0.185
Am Tropical-Monsoon	0.226
Dwa Cold-Dry_Winter-Hot_Summer	0.221
BSk Arid-Steppe-Cold	0.182
Dwb Cold-Dry_Winter-Warm_Summer	0.220
Dfa Cold-Withouth_dry_season-Hot_Summer	0.213
Dfb Cold-Withouth_dry_season-Warm_Summer	0.225
Csb Temperate-Dry_Summer-Warm_Summer	0.195
Dfc Cold-Withouth_dry_season-Cold_Summer	0.220
Cfc Temperate-Withouth_dry_season-Cold_Summer	0.210
BWk Arid-Desert-Cold	0.152
Cfa Temperate-Withouth_dry_season-Hot_Summer	0.222
Cfb Temperate-Withouth_dry_season-Warm_Summer	0.207
Csa Temperate-Dry_Summer-Hot_Summer	0.173
Dwc Cold-Dry_Winter-Cold_Summer	0.219
Dsb Cold-Dry_Summer-Warm_Summer	0.213
Cwa Temperate-Dry_Winter-Hot_Summer	0.210
Dsc Cold-Dry_Summer-Cold_Summer	0.210
Cwb Temperate-Dry_Winter-Warm_Summer	0.192
Cwc Temperate-Dry_Winter-Cold_Summer	0.126
Dfa Cold-Withouth_dry_season-Very_Cold_Winter	0.218
Dsd Cold-Dry_Summer-Very_Cold_Winter	0.213
Dwd Cold-Dry_Winter-Very_Cold_Winter	0.190

larger terrain fluctuations. Jung's further research also quantified the error of wind power density and capacity factor. In this study, however, the spatial and temporal variation of α is analyzed in more detail, and the monthly variation of α is calculated by distinguishing between land and ocean, and latitudes. We also analyzed the diurnal and interannual variation of α values separately from land and ocean.

It is consistent with previous findings that the ocean α value is about 0.11 under near-neutral stable conditions, deduced from several site data and confirmed by other offshore wind potential research [51,52]. Our α results have smaller values on the coastal areas compared to wind tower results in another study [21]. The clear differences and opposite monthly changes in α values between land and ocean are in agreement with previous studies [26]. In our study, α values on land are smaller in summer and larger in winter, this feature also appears in many studies based on regional measurements [14,21,22,26,27,41]. In some studies, the diurnal variation of α values is characterized by being smaller during the day and larger at night. This phenomenon also appears in our study, and we find that such characteristic is more significant on land than in

the ocean at a global scale [15,20,26,27,41]. Similar to previous studies, our results also show that the α value is smaller in unstable conditions (such as mountains) than in stable conditions (such as plains) [19,26]. Regarding the discussion of terrain in this study, some research use model simulations and lidar measurements to focus on regional terrain. They show that some complex terrains may cause α value anomalies, and our study find that such anomalies are common around the globe [20,41]. Generally, at the global scale we find that all these features are common across the world's land and ocean. Evaluation from this study and other relevant studies shows regional variation in the α value that may be caused by latitude, surface roughness, and topography [43,53]. The turbulence intensity and seasonal monsoons may also affect the results diversely in the land and ocean areas [54]. Research we refer to discusses the difference of α value at different wind speeds, which we do not discuss but is worth studying in wind energy applications [23].

Since many related studies in the past are based on regional wind towers, most of their measured wind speed data are within three years and rarely over 20 years [15,19,20,22,26,27]. Compared with other studies using the data at a short temporal scale [21], the multi-year monthly, annual, and interannual analyses of the 43-year hourly data rather than several days or years make the characteristics of the α value more accurate and comprehensive. In addition, we found that the STD values are relatively high on land with greater increasing elevation and mountainous terrain, which has a similar result as the previous research [32]. As for wind energy assessment, the trends of global wind speed changes in recent decades are not significant, but the trend of α value cannot be neglected [55,56]. Most of the related studies in the past are conducted using wind towers or lidar measurements. These studies focus on one region and have spatial limitations. Some studies use measured data to verify model simulation values, but only two representative stations that are close to each other are verified [20]. Our analysis not only ensures the reliability of the data results, but also gives the results a spatial breadth. This is something that many previous studies have failed to do using wind towers, radar, remote sensing data, or reanalysis data alone.

4.2. Limitations of the ERA5 results

Such results, however, still have certain limitations in application. Since each grid in the ERA5 reanalysis data covers a certain spatial range, it is not an exact location. Therefore, if the wind energy output at a specific location is considered, the result calculated using the α value of ERA5 will not be as accurate as the one using a wind tower data especially in coastal and mountainous terrain [49,57]. Nevertheless, wind power calculations at a specific location are only required when wind turbine companies are considering the precise location at which the turbine will be built. Usually, relevant authorities need to conduct a comprehensive assessment of a region's wind resources, which is the

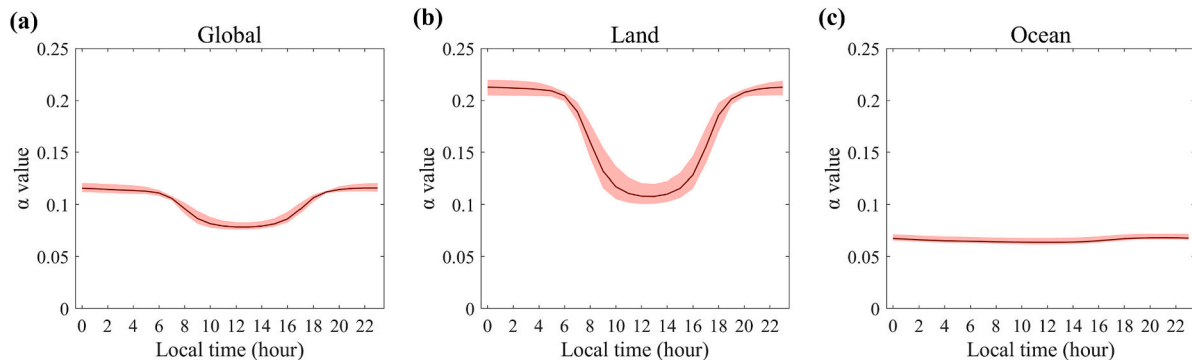


Fig. 6. Diurnal variation of α value. Global land raster data is used to separate the land and ocean areas. The local time is based on UTC time zone. Buffer areas refer to the 12 interannual monthly mean value in every hour during 1980–2022. (a) Global averaged α value in each hour. (b) Global land averaged α value in each hour. (c) Global ocean averaged α value in each hour.

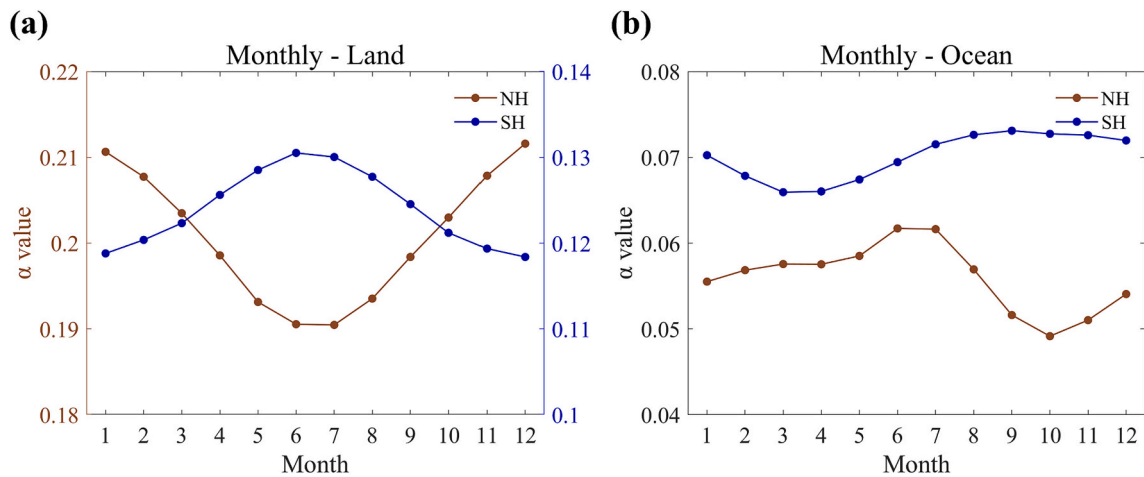


Fig. 7. Seasonal variation of α value in land and ocean in the Northern and Southern Hemispheres. (a) is the land mean α value in North and South Hemisphere differentiated over the 12 months. (b) is the ocean mean α value in Northern and Southern Hemisphere differentiated over the 12 months.

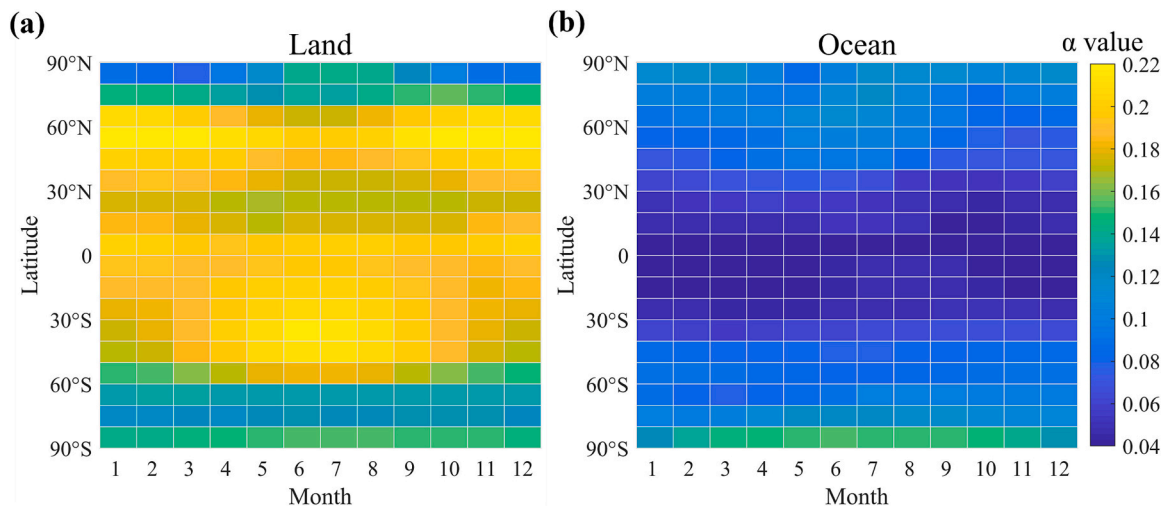


Fig. 8. The seasonal and latitudinal variation of α value. Global land raster data is used to separate the land and ocean areas.

situation for which the global results of this study apply. As the temporal and spatial resolution of such global reanalyzed wind speed data improves, the α results associated with this study will also become more accurate especially in complex terrain.

When calculating α , the ERA5 α is calculated directly using the two heights of 10 m and 100 m, which will bring errors. These errors arise from variation in the height ranges utilized and variation in the number and intervals of adjacent heights employed (a challenge inherent in tower α as well). A study investigated α values within three height ranges spanning 10 m to 80 m using a meteorological tower. Different ranges imply discrepancies in the number of heights considered and the height intervals used, especially including the 10 m wind speed. This variation stems from the heightened atmospheric instability near the ground [19]. Due to data constraints, our calculations are limited to two heights. In the absence of such constraints, we would prefer utilizing as many heights as possible to mitigate errors, akin to the approach taken with wind tower data.

In addition, when we use wind tower data to compare and verify the ERA5 results, all wind towers are located in China due to difficulties in obtaining data. This only shows that the ERA5 results are reliable at these locations, not a global validation. In the future, we hope to expand the location of measurement around the world and conduct global verification of α value.

5. Conclusion

In this study, we first verified the characteristics of the power law of the wind profile by using the anemometer wind tower data located in China. Our findings confirmed that, the power law is applicable at different time scales and spatial ranges. And there are obvious differences between the power law exponent and $1/7$. Then we compared the measured data and reanalysis data, which regional verification can serve as a reliability for global research using ERA5 data. This comparison provided compelling evidence regarding the importance and reliability of using ERA5 reanalysis data to analyze the power law exponent α . Afterwards, we used reanalysis data from ERA5 to determine the global power law exponent, after validating the power law with measured data obtained from wind towers. This approach allowed us to establish a comprehensive characterization of the power law exponent on a global scale. On the temporal scale, we distinguished the diurnal, monthly and annual average data, and then analyzed the average value, STD and trend of the power law exponent α . On the spatial scale, we categorized the data into land, ocean, and different latitudes for a comprehensive analysis of the global characteristic of α .

The α value exhibits varying degrees of spatiotemporal variation across different scales of analysis scenarios. Generally, α values on land are mainly larger than $1/7$, while values in the ocean are generally

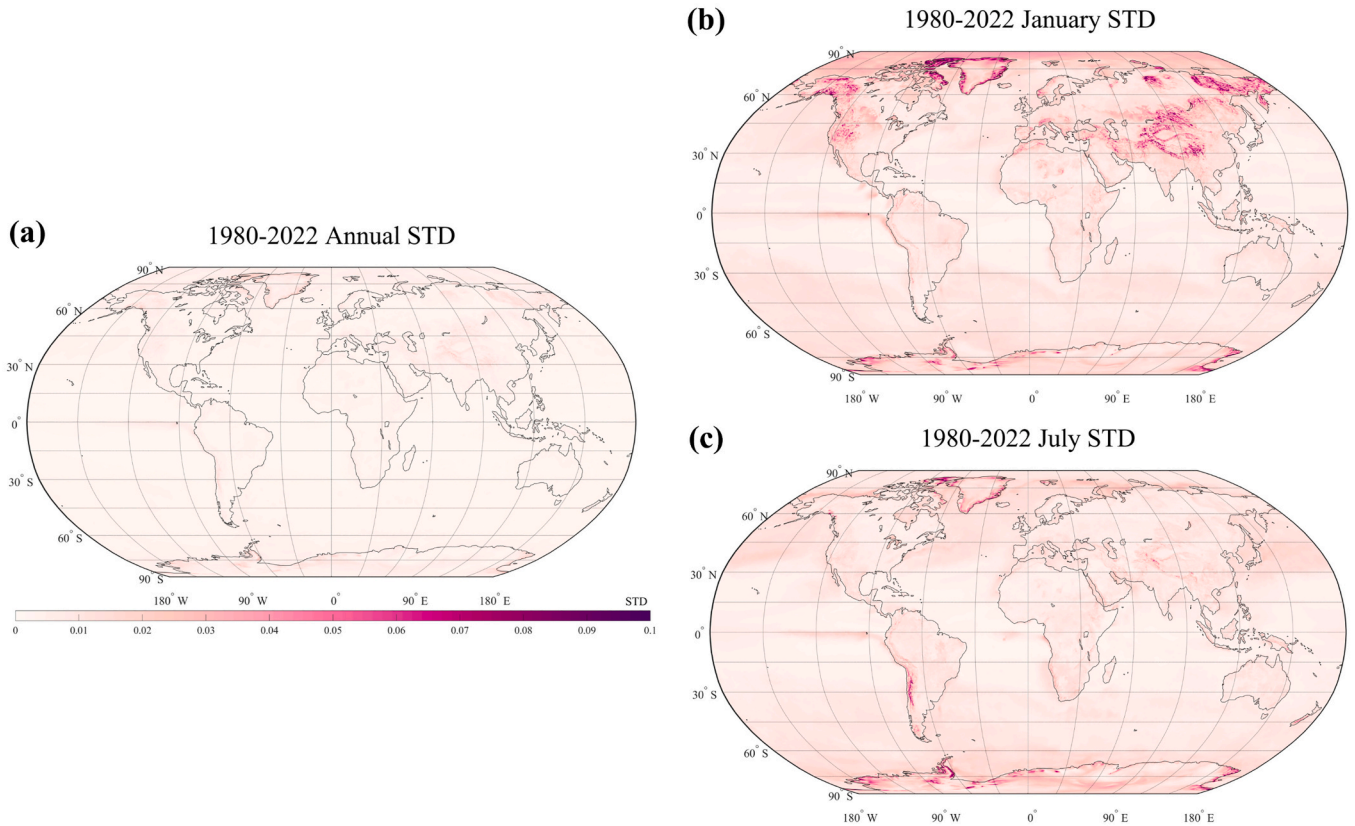


Fig. 9. Spatial pattern of the standard variation (STD) of α value. (a) STD of α value of the 43 years. (b) STD of α value in January of the 43 years. (c) STD of α value in July of 43 years.

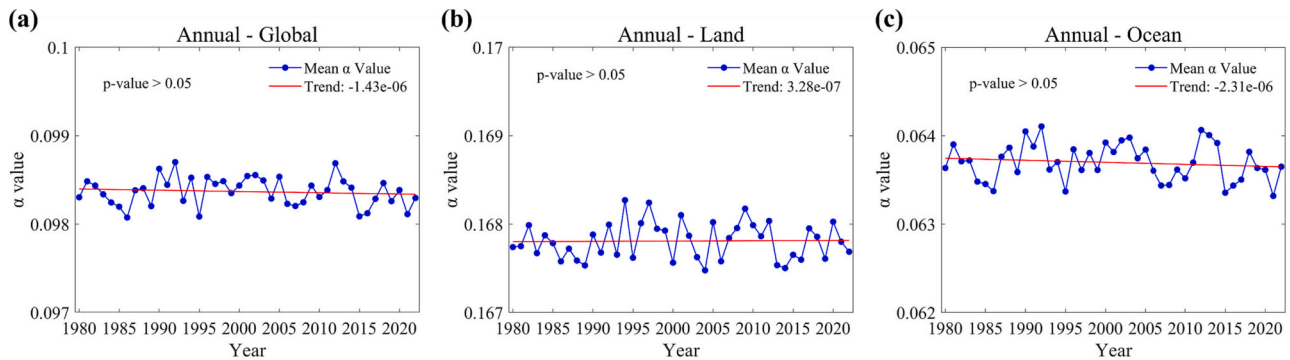


Fig. 10. Time series of α value. (a)-(c) are the interannual average α value data of the globe, land, and ocean areas. Red lines show their trends fitted by least square linear regression, and p -value based on t -test. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

smaller than $1/7$. The results illustrate a gradual decrease in the α value from the equator to the poles on land, while in the ocean, it exhibits an increase. Additionally, the α value shows significant diurnal variation as it is lower in the daytime while higher at night on land. In contrast, the α value remains relatively stable in the ocean throughout the day. From the average seasonal fluctuation, the α value on land tends to increase from summer to winter and then decrease from winter to summer. Conversely, over the ocean, the variation pattern is reversed. For the STD values, the results are relevant to the terrains which vary with regions. Globally, areas with increasing and decreasing trends appear staggered, although most of them are not statistically significant. These observations suggest that diurnal, seasonal, and spatial variation provide more meaningful insights than year-to-year changes in the α value data.

Our results show significant spatiotemporal variation in the α value across different regions of the globe. Furthermore, the α value obtained from the measured data also differs from the uniformly used $1/7$ and reanalysis data to varying degrees at specific locations. This highlights the importance of focusing on local regions when studying wind energy production suitability in future research endeavors. For example, it is worth analyzing most parts of North China, some coastal areas in East China, and part of the Qinghai-Xizang Plateau, which are abundant in wind energy resources [58]. According to our results in these areas, the significant decrease of the α value and increase of the STD value will affect the evaluation of wind energy before the construction of wind turbines. Therefore, careful consideration of these variation is crucial to ensure accurate evaluations of wind energy potential in these regions. In areas with high α values on land at low latitudes, wind speeds may be

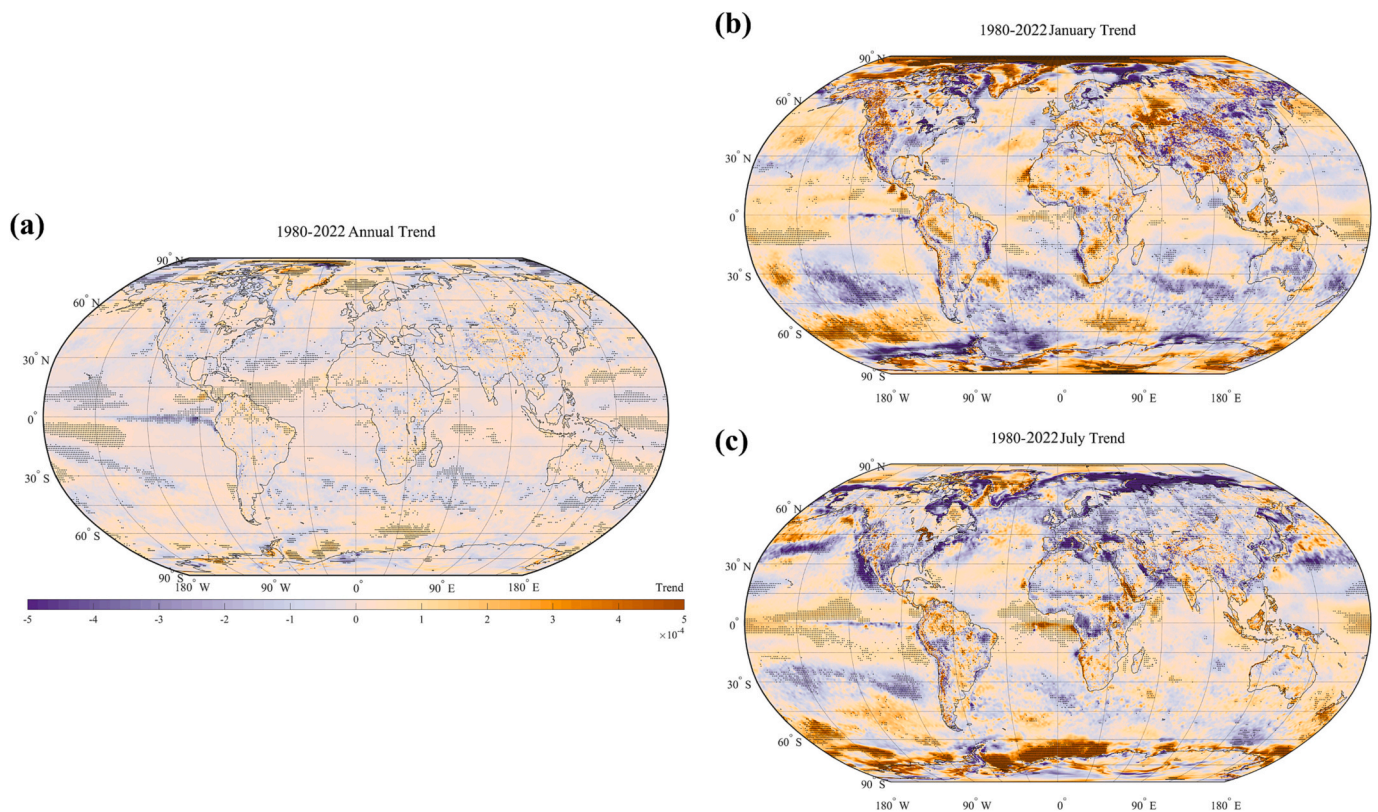


Fig. 11. Spatial pattern of the trend in α value. (a) The trend of α value of the 43 years. (b) The trend of α value in January of the 43 years. (c) The trend of α value in July of the 43 years. The black dot shows the least squares linear fitted trend's p -value less than 0.05 that based on t-test, defined as significant.

low at lower heights but increase significantly at higher heights. This characteristic suggests that such regions could be suitable for wind energy development, considering the potential for harnessing stronger winds at greater heights. Another example is the value of α in offshore is smaller than that on land while the wind speed offshore is larger than onshore that has abundant wind energy [59,60], which may affect the evaluation of the optimal construction height of offshore wind turbines. In short, using global reanalysis data for a large-scale assessment facilitates a comprehensive approach to wind energy utilization research. The subsequent study will also conduct a more detailed assessment of local areas that need wind energy development, such as the influence of different topography on wind speed variation [15].

Such global-scale α value study can provide a better α reference value for using power law to obtain upper-level wind speed anywhere in the world, which cannot be provided by previous regional studies. At the same time, our improved estimations of the power law exponent α can be applied by renewable energy developers to assess wind energy potential at specific locations more accurately. More precise calculations of wind turbine energy production can reduce uncertainties in feasibility studies and allow optimized design of wind farms. Moreover, climatologists and atmospheric scientists can gain new insights into boundary layer dynamics from the global maps and long-term α . The spatiotemporal variation reflect impacts of large-scale modes of climate variability and change, as well as regional weather systems and circulation patterns. The reanalysis-based α dataset provides a unique perspective for studies on the effects of climate change on near-surface wind fields on a large spatial scale. To facilitate subsequent wind energy-related research, we will make the data of the global α value with temporal and spatial variation calculated by ERA5 public, and readers can access the data in the following doi: [10.57760/sciencedb.10067](https://doi.org/10.57760/sciencedb.10067).

CRediT authorship contribution statement

Xinrong Yang: Writing – review & editing, Writing – original draft, Methodology, Investigation. **Xin Jiang:** Writing – original draft, Methodology. **Shijing Liang:** Writing – review & editing, Writing – original draft, Methodology. **Yingzuo Qin:** Resources, Methodology. **Fan Ye:** Resources, Funding acquisition. **Bin Ye:** Writing – review & editing, Methodology. **Jiayu Xu:** Writing – review & editing. **Xinyue He:** Writing – review & editing. **Jie Wu:** Writing – review & editing. **Tianyun Dong:** Writing – review & editing. **Xitian Cai:** Writing – review & editing, Supervision, Methodology. **Rongrong Xu:** Writing – review & editing, Resources, Methodology. **Zhenzhong Zeng:** Writing – review & editing, Writing – original draft, Resources, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apenergy.2023.122441>.

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